Genelleştirilmiş Fourier Serisi Açılımı

Hatırlatma

The signals $\psi_1(t), \psi_2(t), \dots, \psi_N(t)$ ortagonal over $[t_1, t_2]$ if

$$\int_{t_1}^{t_2} \psi_m(t)\, \psi_n^*(t)\, dt = \begin{cases} 0, & m \neq n \\ E_m, & m = n \end{cases}$$

In addition, if $E_m = 1$ $\forall m$ the signals are orthonormal.

Herhangi bir $x(t)$ işaretinin bu N ortogonal (olik) işaret ile yaklaşımını elde edelim

$$x(t) \cong c_1\psi_1(t) + c_2\psi_2(t) + \dots + c_N\psi_N(t)$$

$$= \sum_{n=1}^{N} c_n \psi_n(t), \quad t_1 \le t \le t_2$$

Hata işareti (error signal) $= e(t) = x(t) - \sum_{n=1}^{N} c_n\psi_n(t)$

Hatanın enerjisi $= E_e = \int_{t_1}^{t_2} |e(t)|^2 dt$

(Energy of error)

Eğer yaklaşımın katsayıları $c_n = \frac{1}{E_n} \int_{t_1}^{t_2} x(t)\, \psi_n^*(t)\, dt$; $n = 1, \dots, N$ (→ korelasyon) olarak seçilirse hatanın enerjisi minimum olur.

Ortagonal işaret kümesi tam (complete) ise hatanın enerjisi sıfıra gider ve yaklaşıklık eşitlik olur.

$$x(t) = c_1\psi_1(t) + c_2\psi_2(t) + \dots + c_N\psi_N(t) + \dots = \sum_{n=1}^{\infty} c_n\psi_n(t), \quad t_1 \le t \le t_2$$

↳ Genelleştirilmiş F.S açılımı (Generelized FS expansion)

Eğer yaklaşımın katsayıları $C_n = \frac{1}{E_n}\int_{t_1}^{t_2} x(t)\psi_n^*(t)\,dt$; $n=1,\dots,N$

olarak seçilirse hatanın enerjisi minimum olur

Ortogonal işaret kümesi tam (complete) ise hatanın enerjisi sıfıra gider ve yaklaşıklık eşitlik olur:

$$x(t) = C_1\psi_1(t) + C_2\psi_2(t) + \dots + C_N\psi_N(t) + \dots = \sum_{n=1}^{\infty} C_n\psi_n(t), \quad t_1 \le t \le t_2$$

↳ Genelleştirilmiş FS açılımı (Generalized FS expansion)

Basis functions (Temel fonksiyonlar)

($x(t)$ işaret ailesi için $N\to\infty$ $E_e \to 0$ ise $\underbrace{\{\psi_n(t)\}}$ ailesi tamdır. Bu durumda $x(t)$ işaret ailesi içindeki bütün işaretler bu temel fonksiyonlar ile ifade etmek mümkündür.)

Parseval teoremi: $E_x = C_1^2E_1 + C_2^2E_2 + \dots + C_N^2E_N + \dots = \sum_{n=1}^{\infty} C_n^2E_n$

Trigonometrik FS açılımı

$\forall t \quad x(t+T_0) = x(t)$

F. Serisinin mevcut olmasorttan:

1) $x(t)$ bir periyoda boyunca mutlak integrali alınabilirolmalıdır: $\int_{T_0} |x(t)|dt < \infty$

(fourier serisi vardır, her noktada yakınsamaya bilir)

2) $x(t)$'nin bir periyodunda sonlu sayıda maksimumu, minimumu vesüreksizliği vardır.

Zayıf (weak) Dirichlet şartı

Kuvvetli (strong) Dirichlet şartları

İkisi sağlanıyorsa fourier serisi açılımı vardır.

★ Laboratuvarda üretilen işaretlerin fourier açılımı vardır

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n w_0 t) + b_n \sin(n w_0 t)] \qquad t_1 \le t \le t_1 + T_0$$

$$w_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

$$a_0 = \frac{1}{T_0}\int_{T_0} x(t)dt, \quad a_n = \frac{2}{T_0}\int_{T_0} x(t)\cos(n w_0 t)dt \quad n = 1, 2, \dots$$

$$b_n = \frac{2}{T_0}\int_{T_0} x(t)\sin(n w_0 t)dt, \quad n = 1, 2, \dots$$

$x(t)$ tek $\Rightarrow a_0 = a_n = 0$

$x(t)$ çift $\Rightarrow b_n = 0$

$\forall n,m \quad \int_{T_0} \sin(n w_0 t)\cos(m w_0 t)dt = 0$ → ortagonaldir

ortogonal

$$\int_{T_0} \cos(n w_0 t)\cdot\cos(m w_0 t)dt = \begin{cases} 0 & n \neq m \\ T_0/2 & n = m \end{cases}$$

$$\int_{T_0} \sin(n w_0 t)\sin(m w_0 t)dt = \begin{cases} 0, & n \neq m \\ T_0/2 & n = m \end{cases}$$

Parseval teoremi : $P_x = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2)$

Compact FS Expansion

$$a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) = c_n \cos(n\omega_0 t + \theta_n)$$

$$c_n = \sqrt{a_n^2 + b_n^2}, \quad c_0 = a_0$$

$$\theta_n = -\arctan\left(\frac{b_n}{a_n}\right)$$

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cdot \cos(n\omega_0 t + \theta_n) \qquad t_1 \le t \le t_1 + T_0$$

Parseval Teoremi: $P_x = c_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} c_n^2$

Parseval teoremi: $P_x = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2)$

Fourier spektrumu

$c_n \times f = n f_0$ Genlik (Amplitude) spektrumu } Çizgi (Line) spektrumu

$\theta_n \times f = n f_0$ Faz (Phase) // }

Üstel FS Expansion

(Complex/Exponential FS expansion)

$$c_n \cos(n\omega_0 t + \theta_n) = c_n \cdot \frac{1}{2}\left[e^{j(n\omega_0 t + \theta_n)} + e^{-j(n\omega_0 t + \theta_n)}\right]$$

$$= \underbrace{\frac{1}{2} c_n e^{j\theta_n}}_{x_n} \cdot e^{jn\omega_0 t} + \underbrace{\frac{1}{2} c_n \cdot e^{-j\theta_n}}_{x_{-n}} e^{jn\omega_0 t}$$

$$x(t) = \sum_{-\infty}^{\infty} x_n e^{jn\omega_0 t}$$

$$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$\int_{T_0} \underbrace{e^{jm\omega_0 t} \cdot e^{-jn\omega_0 t}}_{e^{j(m-n)\omega_0 t}} dt = \begin{cases} 0, & n \neq m \\ T_0, & n = m \end{cases}$$

Parseval teoremi: $P_x = \sum_{n=-\infty}^{+\infty} |x_n|^2$

$e^{jn\omega_0 t}$ nin gücü 1 dir.

$$x_n = \frac{1}{2} c_n e^{j\theta_n} \quad n = 1, 2, \dots \infty$$

$$x_{-n} = \frac{1}{2} c_n e^{-j\theta_n} \quad n = 1, 2, \dots \infty$$

$$x_0 = c_0 = a_0$$

Üçüncü F. spektrumu

$$x_n = |x_n| e^{j\theta_n}$$

$$x_{-n} = |x_n| e^{-j\theta_n}$$

$$|x_{-n}| = |x_n| = \frac{1}{2} c_n$$

$$\angle x_n = \theta_n, \quad \angle x_{-n} = -\theta_n$$

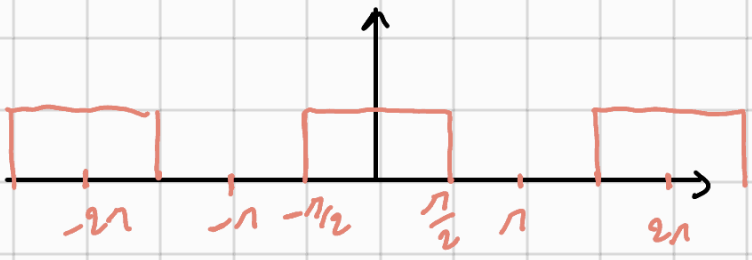
$|x_n| \times f = nf_0$ Genlik spektrumu

$\angle x_n \times f = nf_0$ Faz spektrumu

Örnek: Find compact FS expansion of $x(t)$

(Unipolar square wave)

Tek kutuplu kare dalga



$$\Rightarrow T_0 = 2\pi$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t)\,dt = \frac{1}{2\pi} \int_{-T_0/2}^{T_0/2} x(t)\,dt = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} 1\,dt$$

$\omega_0 = 1$

$$= \frac{1}{2\pi} \pi = 1/2 \text{ V}$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cdot \cos(n \quad t)\,dt \qquad \omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2\pi} = 1$$

$$= \frac{2}{2\pi} \int_{-\pi/2}^{\pi/2} 1 \cdot \cos(n \quad t)\,dt = \frac{1}{\pi} \frac{\sin(n \quad t)}{n} \Big|_{-\pi/2}^{+\pi/2}$$

$$= \frac{1}{\pi n}\left[\sin\left(n\frac{\pi}{2}\right) - \sin\left[-n\frac{\pi}{2}\right]\right]$$

$$a_n = \frac{2}{\pi n}\sin\left(n\frac{\pi}{2}\right) = \begin{cases} 0, & n \text{ çiftse} \\ \frac{2\pi}{n}, & n = 1, 5, 9 \\ -\frac{2\pi}{n}, & n = 3, 7, 11 \end{cases}$$

$x(t)$ çift $\Rightarrow b_n = 0$

ezberle

$$x(t) = \frac{1}{2} + \frac{2}{\pi}\left[\cos(\omega_0 t) - \frac{1}{3}\cos(3\omega_0 t) + \frac{1}{5}\cos(5\omega_0 t) - \frac{1}{7}\cos(7\omega_0 t) + \cdots\right]$$

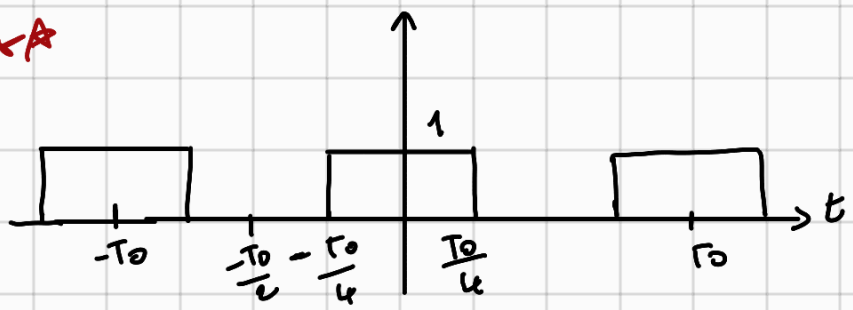
$$-\cos a = \cos(a - \pi)$$

ezberle

$$x(t) = \frac{1}{2} + \frac{2}{\pi}\left[\cos(\omega_0 t) + \frac{1}{3}\cos(3\omega_0 t - \pi) + \frac{1}{5}\cos(5\omega_0 t) + \frac{1}{7}\cos(7\omega_0 t - \pi) + \cdots\right]$$

$$c_0 = \frac{1}{2}, \quad c_n \begin{cases} \frac{2}{\pi n} & n \text{ tek} \\ 0 & n \text{ çift} \end{cases}, \quad a_n = \begin{cases} -\pi, & n = 3, 7, 11, \ldots \\ 0, & n \neq 3, 7, 11, \ldots \end{cases}$$

AA



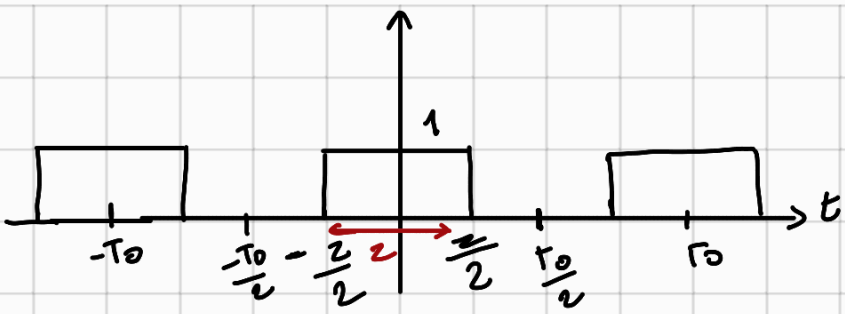
$$a_0 = \frac{1}{2}, \quad a_n = \frac{2}{\pi n}\sin\left(\frac{n\pi}{2}\right), \quad b_n = 0$$

Buradan geçiç

$$a_n = \frac{2}{T_0}\int_{-T_0/4}^{T_0/4} 1\cos(n\omega_0 t)\,dt = \frac{2}{T_0}\frac{\sin(n\omega_0 t)}{n\omega_0}\Big|_{-T_0/4}^{T_0/4}$$

$$= \frac{2}{\frac{n\omega_0 T_0}{2\pi}}\left[\underbrace{\sin\left(n\omega_0\frac{T_0}{4}\right) - \sin\left(-n\omega_0\frac{T_0}{4}\right)}_{2\sin\left(\frac{n\omega_0 T_0}{4}\right)}\right]$$

$$= \frac{2}{\pi n}\sin\left(n\frac{\pi}{2}\right)$$

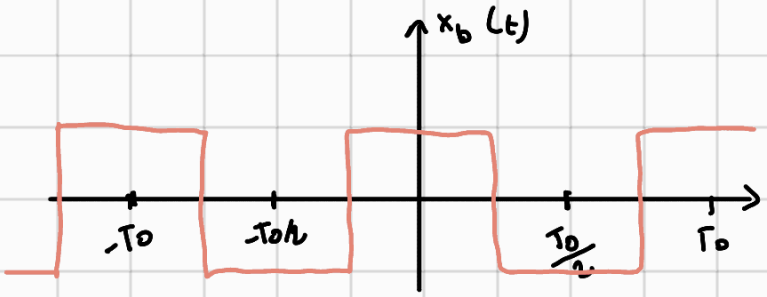


Ödev olarak ispatlayınız

$$a_0 = \frac{\tau}{T_0}, \quad a_n = \frac{2}{\pi n} \sin\left(n \omega_0 \frac{\tau}{2}\right), \quad b_n = 0$$

Bi polar square wave

(Çift kutuplu kare dalga)



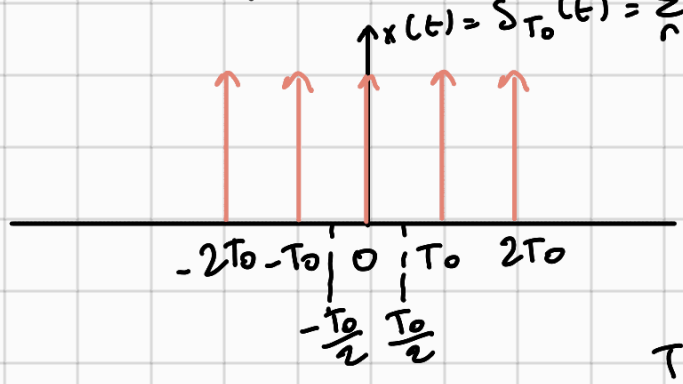
$$x_b(t) = 2x(t) - 1 = 2\left[x(t) - \frac{1}{2}\right]$$

$$x_b(t) = \frac{4}{\pi}\left[\cos(\omega_0 t) - \frac{1}{3}\cos(3\omega_0 t) + \frac{1}{5}\cos(5\omega_0 t) - \frac{1}{7}\cos(7\omega_0 t) + \cdots\right]$$

Örnek Dürtü treni

(Impulse train)

$$x(t) = \delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$



$$a_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \underbrace{x(t)}_{\delta(t)}\,dt = \frac{1}{T_0}$$

$$\int_a^b f(t)\delta(t - t_0)\,dt = \begin{cases} f(t_0) & a < t_0 < b \\ 0 & \text{diğer} \end{cases}$$

$$a_n = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\cos(n\omega_0 t)\,dt = \frac{2}{T_0}$$

$x(t)\cos(n\omega_0 t) \to \delta(t - 0)$

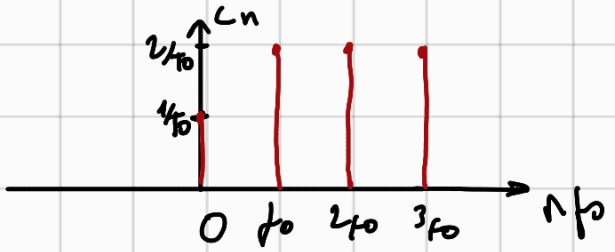
$\cos(n\omega_0 \cdot 0) = 1$

$x(t)$ çift , $b_n = 0$

$$x(t) = \underbrace{\frac{1}{T_0}}_{a_0 = c_0} + \sum_{n=1}^{\infty} \underbrace{\frac{2}{T_0}}_{a_n = c_n}\cos(n\omega_0 t)$$

$$= \frac{1}{T_0}\left[1 + 2\sum_{n=1}^{\infty}\cos(n\omega_0 t)\right]$$

$$c_0 = \frac{1}{T_0}, \quad c_n = \frac{2}{T_0} \qquad \theta_n = 0$$



Üstel FS açılımı

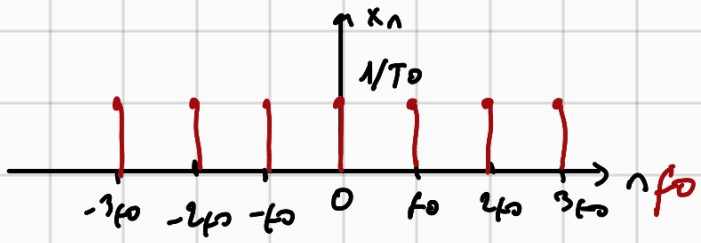
$$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t)\, \underbrace{e^{-jn\omega_0 t}}_{0}\, dt = \frac{1}{T_0}$$

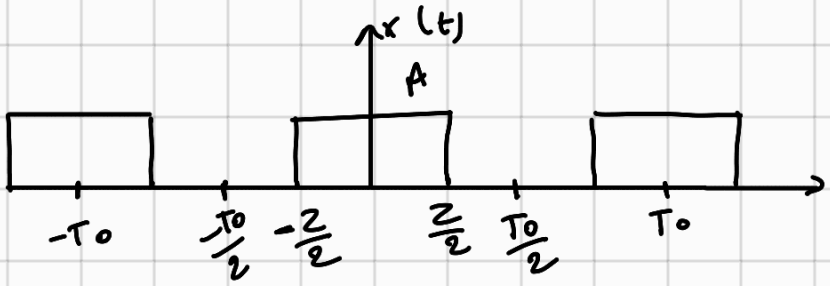
$$x(t) = \sum_{n=-\infty}^{+\infty} x_n e^{jn\omega_0 t} = \frac{1}{T_0}\sum_{n=-\infty}^{+\infty} e^{jn\omega_0 t}$$

$$x(t) = \delta_{T_0}(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT_0) = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

$$x_n = \frac{1}{T_0} \qquad |x_n| = \frac{1}{T_0} \qquad \theta_n = 0$$



Örnek



Üstel FS açılımı

$$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\int_{-Z/2}^{Z/2} A\, e^{-jn\omega_0 t}\, dt = \frac{A}{T_0}\left.\frac{e^{-jn\omega_0 t}}{-jn\omega_0}\right|_{-Z/2}^{Z/2}$$

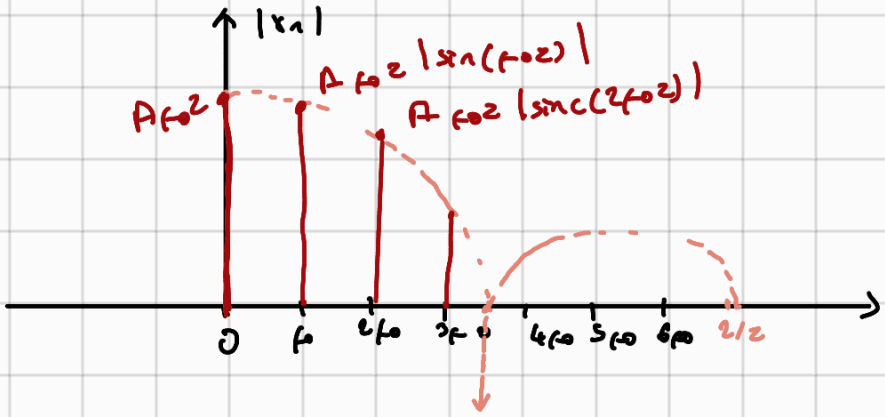
$$= -\frac{A}{jn\omega_0 T_0}\left[e^{-jn\omega_0 \frac{Z}{2}} - e^{+jn\omega_0 \frac{Z}{2}}\right] = \frac{A}{n\pi}\sin\left(n\omega_0 \tfrac{Z}{2}\right)$$

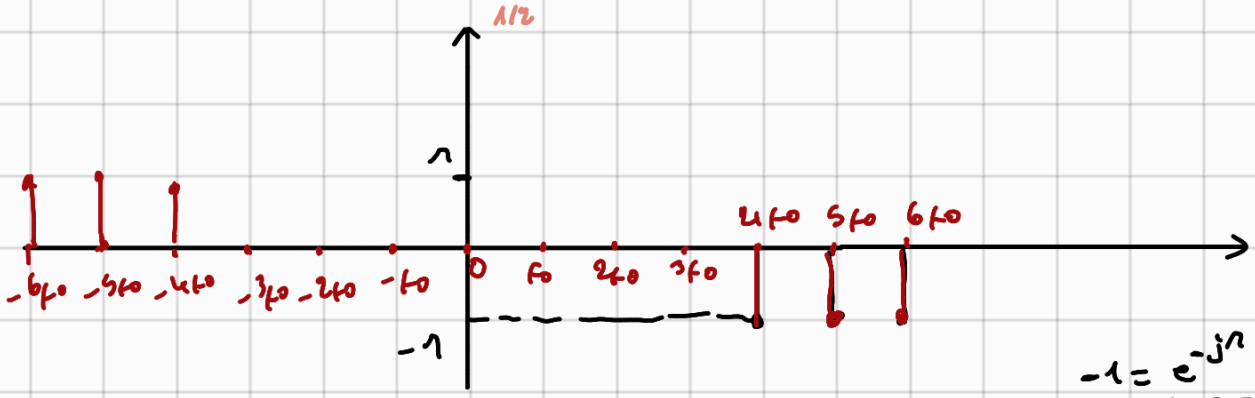
$$x_n = \frac{A}{\pi n}\sin\left(n 2\pi f_0 \frac{z}{2}\right) = \frac{A}{\pi n}\underbrace{\frac{\sin(\pi n f_0 z)}{\pi n f_0 z}}_{\text{sinc}(n f_0 z)} \cdot \pi n f_0 z$$

$$\boxed{x_n = A f_0 z \,\text{sinc}(n f_0 z)}$$

$$x(t) = \sum_{n=-\infty}^{+\infty} x_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{n=\infty} \underbrace{A f_0 z \,\text{sinc}(n f_0 z)}_{x_n} \cdot e^{jn\omega_0 t}$$

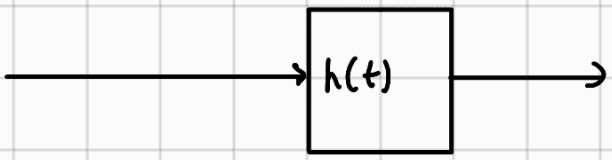
$\frac{T_0}{4} < z < \frac{T_0}{3}$ → $3f_0 < \frac{1}{z} < 4f_0$ aralığında olsun





$$-1 = e^{-j\pi} = \underbrace{\cos\pi}_{-1} - j\underbrace{\sin\pi}_{0}$$

Ödev



$$x(t) = \sum_n x_n e^{jn\omega_0 t} \qquad y(t) = \sum_n y_n e^{jn\omega_0 t}$$

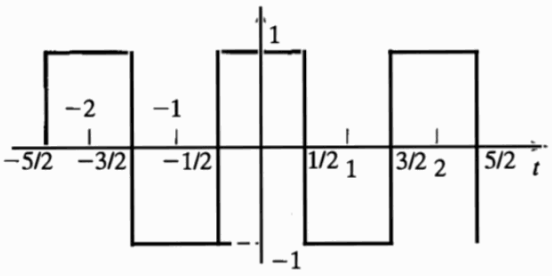
$$y_n = x_n H(nf_0)$$

$$H(f) = \int_{-\infty}^{+\infty} h(t)\cdot e^{-j2\pi ft}\,dt$$

★ Example 2.2.6

Example 2.2.6

Let $x(t)$ denote the signal shown in Figure 2.27, but set the period equal to $T_0 = 10^{-5}$ sec. This signal is passed through a filter with the frequency response depicted in Figure 2.32. Determine the output of the filter.



Figure

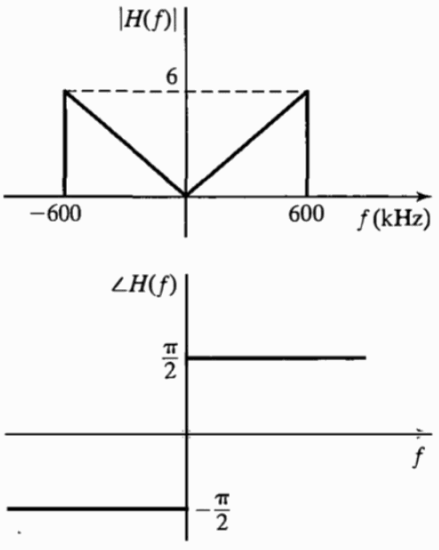


Figure 2.32 Frequency response of the filter.

$$x(t) = \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}\cos\left(2\pi(2n+1)10^5 t\right)$$

$T_0 = 10^{-5}$

$a_0 = 0$ $\quad a_n = \frac{4}{n\pi}\sin(n\omega_0 \tau/2)$ $\quad \omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{10^{-5}} = 2\pi 10^5$

$a_n = \frac{4}{n\pi}\sin\left(n\frac{2\pi}{T_0}\cdot\frac{T_0}{4}\right)$ $\quad \tau = \frac{T_0}{2}$

$= \frac{4}{n\pi}\sin\left(\frac{n\pi}{2}\right)$

$b_n = 0$

$x(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n\cos(n\omega_0 t)\right] + b_n\sin(n\omega_0 t)$

$\sum_{n=1}^{\infty}\frac{4}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cos\left(n\,2\pi 10^5 t\right)$